

DPP 04

CS & IT

Discrete Mathematics

Set Theory & Algebra

Q1 The POSET

($\{2, 3, 5, 30, 60, 120, 180, 360\}; |$) is_____.

- (A) Join semi lattice but not a meet semi lattice
 (B) Not join semi lattice but a meet semi lattice
 (C) A lattice
 (D) Not a semi lattice

Q2 The POSET ($\{2, 3, 4, 6, 12, 18\}; |$) is

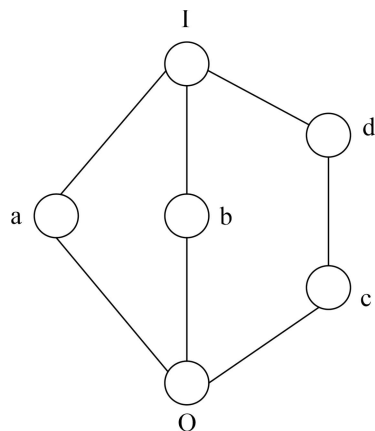
- (A) Join semi lattice but not a meet semi lattice.
 (B) Not join semi lattice but a meet semi lattice.
 (C) A lattice.
 (D) Not a semi lattice.

Q3 Consider the following statements

 S_1 : Every lattice is a totally ordered set. S_2 : Every totally ordered set is a lattice.

- (A) S_1 is true and S_2 is false
 (B) S_1 is false but S_2 is true.
 (C) Both S_1 and S_2 are true.
 (D) Neither S_1 nor S_2 are true.

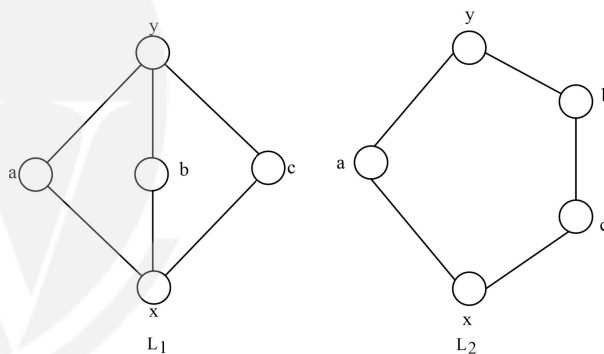
Q4



Which of the following is/are true for above hasse diagram?

- (A) Above hasse diagram represent a complemented lattice.
 (B) Above hasse diagram represent a distributive lattice
 (C) Elements a, b, c, and d have equal number of complements
 (D) Every element of the above lattice has at most one comelement.

Q5

Which of the following is/are true for the lattice 'L' with respect to POSET ($\{1, 2, 5, 3, 9, 90\}, |$)

- (A) L has a sub-lattice which is isomorphic to L_1 .
 (B) L has a sub-lattice which is isomorphic to L_2 .
 (C) L has no sub-lattice which is isomorphic to either L_1 or L_2 .
 (D) L is not a distributive lattice.

Q6 In a Boolean algebra with respect to set A where $|A| = n$ consider the following statements.

S_1 : Number of vertices in the hasse diagram are 2^n .



S2: Number of edges in the hasse diagram are $n \cdot 2^{n-1}$

Which of the following is true?

- (A) S1 is true and S2 is false
- (B) S1 is false but S2 is true.
- (C) Both S1 and S2 are true.
- (D) Both S1 and S2 are false.

Q7 Let P be the partial order defined on the set $\{1,2,3,4\}$ as follows

$$P = \{(x,x) \mid x \in \{1, 2, 3, 4\}\} \cup \{(1, 2), (3, 2), (3, 4)\}$$

The number of total orders on $\{1, 2, 3, 4\}$ that contains P is_____.

Q8 Which of the following is/are always true for any lattice?

- (A) There exists exactly one minimum and exactly one maximum element.
- (B) There exists at most one minimal and at most one maximal element.
- (C) Least upper bound and greatest lower bound exists for every pair of elements.
- (D) Every element has a unique complement.



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Answer Key

Q1 (A)

Q2 (D)

Q3 (B)

Q4 (A, D)

Q5 (A, B, D)

Q6 (C)

Q7 5~5

Q8 (B, C)



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